Updated constants from data analysis

The three Walsh rules recomputed across 719 people in seven cohorts

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What this replaces

An analysis in May tested Walsh's three rules against 138 open-source loop users and found all three wrong, including the parametric form: the slope of log sensitivity on log daily dose came out at -0.43 where the rule implies -1.

This repeats that work on 719 people. The original 138 are included unchanged, so the earlier result can be read beside the new ones rather than replaced by them, and it reproduces exactly.

The main finding is one the smaller cohort could not have shown. **Walsh's sensitivity constant is nearer right for adults than for children and wrong for both**, and the pooled figures everyone quotes, the May analysis included, are driven by which ages happen to be in the sample. In adults it is out by 13% and in children by 41%, in the same direction; the carb ratio rule misses by a fifth at every age; only the basal rule holds.

The three constants

Constant	Walsh	All 719	95% interval	Adults only	Under 18
Sensitivity x daily dose	1700	2108	2048 to 2183	**1915**	2390
Carb ratio x daily dose	500	409	392 to 422	404	414
Basal share of daily dose	0.50	**0.49**	0.48 to 0.50	0.52	0.47

Read across the sensitivity row rather than down it. Both the pooled and the adult columns cover all 719 people; the under-18 column is the JAEB studies alone, because the OpenAPS Commons cohort contains no children. Pooled, the constant sits 24% above Walsh and the interval excludes it. Restricted to adults it lands at 1915, and within that the two sources disagree: 1799 in the JAEB studies against 2381 in Commons, whose members set their own numbers rather than being given them. For JAEB people over 45 it falls to 1766 with an interval of 1680 to 1839, which is the one cell in this entire analysis consistent with 1700.

Half of the pooled sample is under 18, so the headline 2108 is close to an average of two different populations rather than a description of either. That is the reason to read the row and not the corner, and it is the trap the May analysis fell into.

The carb ratio rule is the clearest failure and the most consistent. At 409 it misses 500 by a fifth in every cohort and at every age, so people enter more insulin per gram than the rule prescribes.

The basal rule survives. At 0.49 across everyone the interval covers Walsh's half exactly. That is worth stating plainly, because it is the only one of the three that holds, and it holds despite wide variation underneath it.

Sensitivity by age, which is where the pooled figure comes from

Age band	n	Sensitivity x dose	Interval	Consistent with 1700
2 to 5	115	2163	2045 to 2389	no
6 to 12	128	2561	2406 to 2695	no
13 to 17	54	2350	2243 to 2624	no
18 to 44	162	1830	1738 to 1944	no
45 and over	122	1766	1680 to 1839	**yes**

The constant falls steadily from age six onward and only reaches Walsh's value in the oldest group. A child of ten needs roughly half again the sensitivity factor the rule would give them, which in dosing terms means the rule would correct a child about 50% harder than their own settings say.

The carb ratio moves the other way and more weakly: 297 at ages 2 to 5, rising to 479 by 6 to 12, then settling near 400 in adults. Only the school-age bands are consistent with the 500 rule.

The parametric form, and a caution about how it is usually tested

Walsh's rule implies that sensitivity is inversely proportional to daily dose, so a log-log slope of exactly -1.

Cohort	n	Slope	Interval	Excludes -1
All JAEB pooled	581	-0.98	-1.03 to -0.93	no
JAEB adults only	284	-0.73	-0.82 to -0.65	yes
JAEB under 18	297	-0.91	-0.96 to -0.85	yes
OpenAPS Commons	138	-0.43	-0.59 to -0.27	yes
Everything	719	-0.87	-0.93 to -0.82	yes

The pooled JAEB slope of -0.98 looks like a confirmation of Walsh and is not one. Restricting the same people to adults moves it to -0.73. What produces the tidy -1 is mixing young children, who use little insulin and have high sensitivity factors, with adults who do neither. Between-group differences in level create a gradient that has nothing to do with how sensitivity varies within anybody.

This is not a range effect. The v7 and Loop cohorts span an identical spread of daily doses, 1.43 in logs, and return slopes of -0.46 and -0.94.

So a validation of the 1700 rule performed on a mixed-age cohort will tend to confirm it, and the confirmation will be an artefact of composition. That is the methodological point of this analysis and it applies to the May result as much as to anyone else's.

Results by cohort

Cohort	n	Sensitivity x dose	Carb ratio x dose	Basal share	Slope
Loop	194	2043 (1913 to 2249)	375 (357 to 407)	0.58	-0.94
REPLACE-BG	192	1853 (1791 to 1936)	431 (410 to 460)	0.51	-0.72
DCLP5	100	2429 (2306 to 2580)	513 (479 to 542)	0.46	-1.00
PEDAP	95	2162 (2045 to 2317)	291 (266 to 317)	0.41	-0.89
v5 Trio	22	2682 (2216 to 2962)	373 (279 to 459)	0.45	-0.53
v6 AAPS classic	19	2953 (2201 to 5713)	526 (409 to 607)	0.39	-0.03
v7 oref0	97	2164 (2044 to 2566)	387 (315 to 436)	0.48	-0.46

DCLP3 and IOBP2 are absent because neither release ships a settings file. The bionic pancreas has no sensitivity factor to record by design.

Three cohort-level observations.

The paediatric cohorts differ from each other more than from the adults. PEDAP, ages 2 to 5, has the lowest carb ratio constant at 291 and the lowest basal share at 0.41. DCLP5, ages 6 to 13, has the highest sensitivity constant among the paediatric groups at 2429 and a carb ratio constant of 513 that is consistent with Walsh.

Loop's basal share of 0.58 is the highest here and the furthest from Walsh. It is also a delivered share rather than a programmed one, and under an automated system those differ, so it is not directly comparable to the OpenAPS Commons figures, which are programmed.

The three OpenAPS Commons cohorts all have shallow slopes, from -0.03 to -0.53, while the JAEB cohorts sit between -0.72 and -1.00. These are people who have tuned settings themselves over years against people whose settings came from a clinic within a trial. A rule fits best where people follow it.

Measured rather than entered

Everything above multiplies a setting somebody typed into a pump by their daily dose, which answers what people run. It does not answer what the constant should be. Two of the three rules can be measured from behaviour instead.

Carb ratio: 415, and the measurement is trustworthy

For a meal that was announced, dosed for, and left glucose where it started, the ratio that person needed was the grams divided by the units. No insulin model, no basal assumption, no settings file.

Of 78,705 announced meals from 1,011 people, 23,596 ended within 20 mg/dL of their starting glucose. Restricting to people with at least eight such meals gives 742.

Cohort	n	Carb ratio x dose	95% interval	Covers 500
Loop	594	418	408 to 429	no
REPLACE-BG	148	390	362 to 427	no
Everyone	742	**415**	405 to 425	no

The estimator passes its own honesty test. Across the 289 people who also have an entered carb ratio, entered divided by measured is **1.02**. So this measurement carries no meaningful bias, and 415 is what the rule should be rather than what people happen to have typed. It agrees with the entered figure of 409 because in this case people have already found the right answer.

Basal share: 0.51, but it belongs to the system rather than the person

Delivered basal over delivered total needs no model at all, so it covers the two cohorts with no settings file. That is 438 people every table above omits.

Cohort	System	n	Basal share	Covers 0.50
Loop	DIY loop	830	0.60	no
REPLACE-BG	none	196	0.51	yes
DCLP3	Control-IQ	112	0.48	yes
DCLP5	Control-IQ	100	0.46	no
PEDAP	Control-IQ	95	0.41	no
IOBP2	bionic pancreas	326	0.39	no
Everyone	mixed	1,659	0.51	no

Restricting to adults removes age as the explanation and the pattern sharpens. REPLACE-BG, where no algorithm is running, sits at 0.51 and Control-IQ at 0.50, both on Walsh. The bionic pancreas delivers 0.40 and the do-it-yourself loop 0.63.

So Walsh's half describes what people programme when nothing is adjusting it, and survives one automated system unchanged. The other two move it substantially, in opposite directions, and that is a property of the algorithm rather than of the person using it.

Sensitivity: what a unit does, and why that is not the constant

The third rule resists the treatment that worked for the other two, and it took the whole cohort to see why.

The design is the overnight window used throughout INV-009. Carb-free, quiet before it opens, and the fall compared against nights from the same person at the same hour, the same starting glucose, the same slope into the window and the same level an hour earlier. The exposure is insulin delivered above or below that person's own scheduled basal, by whatever route it arrived, because an algorithm running high temporary basal has given a correction and one that suspends has given a negative one.

Study	control	events	matched fall, 1-3.5 U	per acting unit	entered
REPLACE-BG	open loop	239	22 to 35 mg/dL	**15 to 20**	45
Loop	DIY closed loop	269	-4 to 5	-2 to 5	57
DCLP3	Control-IQ	79	-8 to -4	-5 to -4	-
DCLP5	Control-IQ, 6-13y	38	-7	-9	68
PEDAP	Control-IQ, 2-5y	50	-6	-11	161

Only the open-loop cohort returns anything. The reason sits in the control nights rather than the treated ones. In REPLACE-BG a night with no correction falls 18 to 25 mg/dL over four hours; under an algorithm the same night falls 51 to 59 in Loop, and in PEDAP as much as 146 from the highest starting glucose. Those nights were never untreated. The algorithm corrected them through basal and arrived in the same place, so an added bolus has nothing left to do and nothing to measure.

That is not a limitation of the window design, and conditioning harder does not help. Tightening the match from glucose alone to glucose, slope and the level an hour back moves the Loop estimate from 2.8 to 2.6 mg/dL per unit, a plateau, which rules out error in the reconstructed dose. What remains is that the controller's dose is tied to need the CGM trace does not show: within an identical trace, the correlation between insulin committed before a window and insulin given during it is 0.054 in REPLACE-BG and 0.447 in Loop. Someone boluses once and goes to sleep, while a controller sampling every five minutes keeps answering something we cannot see. No amount of conditioning on the trace breaks a tie to what is not in it.

In the one cohort where the question can be asked, a unit does 15 to 20 mg/dL overnight against 45 entered. As a constant that is 630 to 840 against 1886. The ratio of measured to entered is 0.39 to 0.50 and it holds across four studies, reproducing the affine relation INV-009 found on oref data with a different cohort and a different estimator.

Three things could produce a ratio near 0.4 and this design cannot separate them: a four-hour window truncating a six-hour insulin tail, people correcting precisely when they expect glucose to stay up, and entered settings being optimistic. Extending the horizon would settle the first.

What does survive is the shape, because a common multiplier cannot change a slope. The measured constant does not drift with daily dose: slope +0.106, interval -0.026 to +0.241. A single constant does fit across the dose range, which is the part of Walsh's claim this data settles. What that constant equals, it does not.

The practical reading is that these are two quantities and not two estimates of one. A sensitivity factor in a pump is a dosing parameter, and it encodes the whole fall a person expects after a correction, including the part that would have happened anyway. The measurement above deliberately removes that part. Replacing one with the other would multiply correction doses by about two and a half, which is why the entered constant remains the number to set and the measured one remains the number that describes physiology.

What to do with this

For an adult, the measured figure is 1915 divided by daily dose and the interval, 1828 to 1986, excludes 1700. Walsh survives only in the JAEB over-45s at 1766. So 1700 is a reasonable floor rather than the centre: it will read slightly strong for most adults, which is the safer direction to be wrong in for a starting number, but it should not be defended as the measured value. For a child it is too aggressive by roughly a third, and the gap widens through the school-age years.

The 500 rule should be about 415. That comes from measurement rather than from settings and carries no detectable bias, so it is the firmest number in this document.

The half-of-daily-dose basal convention can stay for someone with no automation and for Control-IQ. It does not describe what a do-it-yourself loop delivers, at 0.63 in adults, or a bionic pancreas at 0.40, and neither of those is a setting anybody chose.

None of these are settings to hold to. INV-009 measured the spread of what people actually run against the rule at 0.68 to 2.07, so any constant is a starting point that then needs individual adjustment, and a quarter of people end up beyond what an automated system's own adjustment range could reach from it.

Method

Settings are those people had entered, taken as a per-person median: from the bolus calculator for Loop and REPLACE-BG, and from the case report forms for DCLP5 and PEDAP. Daily dose is measured from each person's own pump record over days the pump was recording completely, requiring at least thirty. The OpenAPS Commons figures are the May extraction unchanged.

Intervals are percentile bootstrap over people, 4,000 resamples. Slopes are least squares on the logarithms with 1,500 bootstrap resamples.

Basal share is the programmed share where a study records a schedule and the delivered share otherwise, and the two are labelled separately in the machine readable output because they are not the same quantity.

Measured carb ratio takes announced meals of at least 20 g with a dose within 20 minutes, no other carbohydrate for four hours before or five hours after, and no further insulin during, keeping those that ended within 20 mg/dL of their starting glucose. Meals that did not end neutral are excluded rather than corrected.

Code is `inv009/walsh_constants.py` for the entered figures and `inv009/measured_basal_cr.py` for the measured ones; `inv009/measured_constant.py` holds the sensitivity attempt and documents why its level is not reported. Results are in `results/inv009_walsh_by_cohort.parquet`, `results/inv009_measured_basal_cr.json` and their companions.